

DHANUSH

AI Engineer

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Summary

AI Engineer building production-ready GenAI applications using LLMs, RAG, FastAPI, and LangChain. Experienced in scalable backend systems, AI automation workflows, and API integrations. Passionate about shipping real-world AI products with measurable impact.

Education

B.Tech. in Mechanical Engineering

Indian Institute of Information Technology, Design and Manufacturing, Chennai, India

Expected 2026

CGPA: 9.1 / 10

Skills

Programming: Python, SQL, JavaScript

AI & Machine Learning: LLMs, RAG, NLP, Sentiment Analysis, Deep Learning, Prompt Engineering, Supervised Learning

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, LangGraph, Sentence Transformers

Backend & APIs: FastAPI, Flask, REST APIs, Streamlit, Backend Systems, API Integrations

MLOps & Deployment: Docker, Git, GitHub, CI/CD, MLflow, Linux Basics, Model Monitoring

Vector Databases & Retrieval: FAISS, Semantic Search, Embedding Pipelines

Developer & AI Tools: Cursor, Claude Code, Copilot, Zapier MCP, Groq Cloud

Cloud & Databases: AWS, SQLite, ETL Pipelines, Workflow Automation

Experience

AI Engineering Intern

Jan 2026 – Present

Renault Nissan Technology & Business Centre India

Tools Used: Python, FastAPI, Docker, PyTorch

- Developed and deployed AI-driven backend systems using FastAPI and Docker, improving workflow automation efficiency by **45%**.
- Built scalable ML orchestration pipelines integrating APIs, predictive models, and monitoring systems ensuring production reliability and low-latency inference.
- Optimized internal automation workflows and reduced operational processing time by **38%** through AI-assisted backend services and intelligent decision pipelines.

AI Engineering Intern

May 2025 – July 2025

Tata Consultancy Services – Research & Innovation

Tools Used: Python, FastAPI, LangGraph, PostgreSQL

- Built an agentic AI assistant for enterprise incident workflows reducing manual debugging and resolution effort by **40%** through automated issue analysis and remediation generation.
- Developed multi-agent orchestration pipelines using LangGraph improving workflow execution accuracy by **35%** via contextual reasoning, task planning, and intelligent tool-calling.
- Engineered scalable FastAPI backend services supporting low-latency AI automation and reducing operational response time by **45%** across internal enterprise workflows.

Projects

Autonomous AI Invoice Processing & Finance Automation Agent | Claude Code, MCP Servers, Zapier, Python2026

- Built and deployed an end-to-end AI finance automation agent that processed invoices directly from Gmail, extracted structured data, validated vendor records, and automated accounting workflows using **Claude Code, MCP Servers, and Zapier**.
- Integrated Zapier MCP architecture with **8,000+ applications**, enabling scalable enterprise-grade workflow automation without brittle custom integrations.
- Designed intelligent verification pipelines for fraud validation, approval automation, QuickBooks integration, and real-time Slack alerts reducing manual invoice processing effort by **60%**.

RecoverAI Copilot — RAG-Powered Financial Conversation Assistant | *FastAPI, FAISS, Streamlit, Llama 3* 2026

- Engineered a production-ready conversational AI assistant using **RAG**, **FAISS**, **FastAPI**, and **Llama 3** for compliant, sentiment-aware financial workflows.
- Built semantic retrieval pipelines using Sentence Transformers and FAISS enabling context-aware response generation with improved compliance consistency and reduced hallucination risks.
- Developed scalable FastAPI backend services and Streamlit frontend workflows with real-time sentiment analysis and low-latency conversational inference.

Agentic AI Coder Buddy for Autonomous Web App Generation | *LangChain, LangGraph, GPT-OSS* 2025

- Built an agentic AI coding assistant capable of generating complete web applications from natural language prompts using multi-agent orchestration workflows.
- Designed planner, architect, and coder agent pipelines using **LangGraph** improving autonomous code generation reliability and structured reasoning.
- Integrated Groq Cloud inference pipelines enabling fast application generation and scalable AI-assisted developer workflows.

Multilingual AI Content Generation Platform for LinkedIn Workflows | *Llama 3.2, Streamlit, Python* 2025

- Built a multilingual AI content generation platform using **Llama 3.2**, **LangChain**, and **Streamlit** with customizable tone control and workflow automation.
- Developed prompt orchestration pipelines enabling dynamic style adaptation and structured content generation for professional workflows.
- Deployed an interactive Streamlit-based interface improving usability and real-time AI-assisted content generation workflows.

Production NLP Sentiment Analysis Pipeline Optimization | *Python, NLP, Scikit-learn* 2025

- Optimized production NLP pipelines reducing inference latency while improving memory efficiency and prediction accuracy by **20–30%**.
- Benchmarked multiple sentiment analysis models using reproducible experimentation frameworks and structured evaluation metrics.
- Improved scalability for production NLP systems through efficient preprocessing, optimized inference workflows, and robust testing pipelines.

Open Source Contribution — Scalable Sentiment Analysis Framework | *Python, GitHub, Unit Testing* 2025

- Contributed production-grade improvements to an open-source sentiment analysis framework by implementing testing pipelines, validation handling, and scalable modular workflows.
- Implemented robust unit testing, error handling, and modular code architecture following Python best practices and PEP 8 standards.
- Enhanced maintainability and developer onboarding through detailed documentation, structured README updates, and scalable contribution workflows.

Awards & Certifications

- **Academic Excellence Awards (2024 & 2025):** Gold Medal (2024), Silver Medal (2025).
- **Vashisht Tech Fest – Prompt Engineering (2025):** 1st Place Winner.
- **National Yoga Championship (2014):** Gold Medal Winner.